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Perceiver characteristics and target race influence perceptions of the imposter phenomenon

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Abstract – We sought to better understand adults' perceptions of successful women who varied in race and their experience with the imposter phenomenon. Participants responded to scales to assess their own imposter phenomenon and empathy. Next, we randomly assigned participants to one of four conditions where they read about a highly educated woman working in a male-dominated context. The conditions described the woman as either Black or White. The two "Imposter" conditions described the woman as feeling like an imposter. The other two "Control" conditions described the woman as feeling confident although finding it difficult to work in a male dominated industry. After a manipulation check, we asked participants their perceptions of the woman. Compared to Black participants, White participants reported higher levels of imposter phenomenon. Similarly contradicting previous research, our participants did not assume that Black women in our scenarios were more likely to be experiencing imposter feelings. Our participants were more impressed with the Black women who was succeeding in a male-dominated workplace than with the White women in the same circumstance. Adults who experienced imposter phenomenon themselves were more understanding of Black women, while adults higher in cognitive or affective empathy reported more positive perceptions of the White women.

The imposter phenomenon is the feeling that one's success is fraudulent and one's abilities are being overestimated by others (Bhama et al., 2021; Clark et al., 2021). An intense sense of incompetency and doubt often accompanies these feelings (Narayanamoorthy et al., 2023). People who experience imposter feelings also report fear of failure, fear of being deceptive, concerns about their competency, internal conflict, and external attributions of success (Chakraverty, 2022; Chakraverty & Rishi, 2022; Huecker et al., 2023). Imposter feelings are common, with over 90% of college students reporting moderate or higher levels (Fassi et al., 2020) and prevalence rates at over 50% and higher in graduate students, nurses, and medical students (Bravata et al., 2020a).

Ironically, people with imposter feelings have often reached an elevated level of success in highly competitive environments, such as academia or the medical field (Bhama et al. 2021, Narayanamoorthy et al., 2023). Adults who have achieved formal leadership roles, characterized by high visibility and high levels of responsibility, are more likely to report experiencing imposter feelings (Kark & Peters, 2022). Tefwik (2022) explored potential advantages of imposter feelings and discovered that individuals, who experience these feelings in the workplace, are more likely to be other-focused, resulting in higher evaluative ratings for interpersonal effectiveness than their co-workers.

Although people who experience imposter feelings are often successful, researchers have consistently linked the experience to a range of negative outcomes. Imposter feelings are associated with higher levels of depression, anxiety, compromised educational attainment, and lower self-esteem (Bravata et al., 2020a; Canning et al., 2019; Wyat et al., 2019). The experience of imposter feelings often leads individuals to engage in a cycle of over-preparation or procrastination, driven by a desire for perfectionism (Huecker et al., 2023; Pannhausen et al., 2022). Imposter feelings co-occur with maladaptive perfectionistic behaviors such as dwelling on mistakes, experiencing self-doubt, unrealistic expectations, and fear of failure (Cokley et al., 2018; Sheveleva et al., 2023). Perhaps resulting from these perfectionism-driven behaviors, higher rates of imposter feelings also predict higher rates of burn-out symptomology (Jaremka et al., 2020; Vaa et al., 2023). For example, adults experiencing these feelings report feeling depersonalization and emotional exhaustion, along with experiencing lower levels of personal

accomplishment (Alrayyes et al., 2020; Liu et al., 2022; Wang et al., 2019).

Although both men and women can experience imposter feelings, it occurs at higher rates in women. This gender difference was documented over two decades ago (Clance & Imes, 1978) and is still seen, particularly in industries dominated by men (Collins et al., 2020; Clark & Payne, 2024). Compared to their male counterparts, higher levels of imposter feelings are consistently documented in female adults (Fleischhauer et al., 2021), college students (Pákozdy et al., 2023), graduate students (Cowle et al., 2018), and surgical residents (Narayanamoorthy et al., 2023). When quota systems are used, women may be especially prone to doubt their hiring qualifications and thus, even more susceptible to feeling like an imposter (Wall & Troisi, 2020). College students who associated themselves with negative aspects of femininity were more likely to engage in social comparison and experience greater levels of imposter feelings (Fassi et al., 2020).

Race and ethnicity also impact prevalence rates of imposter feelings, with underrepresented groups more vulnerable to experiencing them (Bravata et al., 2020a; Heslop et al., 2023). One reason is that Black, Indigenous, and other people of color are often underrepresented in highly competitive work and academic settings, leading to a perception of not belonging and, thus, feeling like an imposter (Afran et al., 2020). Being one of few people of color often leads to feeling like a token group member rather than a qualified worker (Wall & Troisi, 2020; Wyatt et al., 2019). For example, ethnically underrepresented medical students were more likely to report imposter feelings when enrolled at a primarily White institution compared to a historically Black college or university (Rice et al., 2023). Cokley et al. (2022) found that stress levels are high among Black attorneys, who represent a small minority (about 5%) of all attorneys in the United States; these stress levels served as a mediator between Black attorneys' imposter feelings and negative health outcomes of stress and depression. Another contributing factor is that minority individuals are more likely to experience microaggressions which creates a sense of feeling like an outsider and predict higher rates of imposter feelings (Chakraverty, 2022). Encounters of discrimination and implicit bias elicit feelings of exclusion and thus, amplify the experience of imposter feelings (Cawcutt, 2021; Jenkins et al., 2022; Meadhbh et al., 2023).

Minority women seem to be especially vulnerable to imposter feelings because of the intersection of two risk factors, gender and race. An investigation of first-generation, minoritized status women attending college revealed that imposter feelings were a common experience (Jackson & Colson-Fearon, 2022). Similarly, female medical students from underrepresented groups reported more imposter feelings than their male counterparts regardless of whether they were at a primarily White or primarily Black educational institution (Rice et al., 2023). In the workplace, women of color's feeling like an imposter are likely exacerbated because of work cultures that implicitly or explicitly promote patriarchal views (Collins et al., 2020; Jenkins et al., 2022).

Although imposter feelings are common in women from marginalized groups, researchers have argued that highly educated Black women are especially prone to this self-perception (Chrousos & Mentis, 2020; Fields & Cunningham-Williams, 2021). In a comparison of Asian American, Latinx, and Black college students, Stone-Sabali et al. (2024) found that the Black students were more vulnerable to imposter feelings because of their greater proneness for shame and anxiety. In addition, Black women are significantly underrepresented in leadership roles, which can facilitate an individual Black woman's perception that she does not belong and is an imposter (Manongsong & Ghosh, 2021).

Much of the previous research has focused on characteristics and outcomes of people experiencing imposter feelings. More recently, researchers have started to investigate how people's perceptions impact the process of labeling. People with implicit bias may unfairly label minority women as being imposters whether it is true or not (Feenstra et al., 2020; McGee et al., 2021). These labeled women are typically high achieving; thus, the mislabeling is not the result of their poor performance (Bahama et al., 2021). Instead, the mislabeling is likely the result of biased perceivers, emphasizing demographic attributes of the target woman, such as race and gender (Cawcutt et al., 2021; Simons, 2021). Such labeling may create a self-fulfilling prophecy, because women who perceive themselves to be the subject of negative stigmas and stereotypes in the work force are more likely to report having imposter feelings (Collins et al., 2020). For example, among female software engineers, an increased awareness of gender stigma was associated with increased experiences of imposter feelings (Maji & Dixit, 2023). Ironically, Chakraverty and Rishi (2022) found that while women in certain STEM disciplines were more likely to experience imposter feelings than men, the perpetuators who elicited those feelings included both women and men.

Because imposter feelings are so prevalent, researchers have worked to identify factors that may

decrease the experience, particularly for Black women. Manongsong and Ghosh (2021) argued that a network of supportive others can enable minoritized women to develop a positive leader identity. Another intervention strategy focused on helping women identify the presence and consequences of imposter feelings, with awareness of those feelings minimizing its impact (Parker et al., 2023). These interventions have an underpinning element of empathy and understanding, leading us to wonder whether empathy might reduce attributions of imposter feelings.

Our study focused on attributions of imposter feelings in the workplace, varying in whether the symptoms were explicitly framed as imposter feelings or not and whether the person experiencing it was a Black or White woman. We also investigated how the perceiver's race, empathy level, and personal imposter feelings impacted their judgments of the women in the imposter feelings situation. We hypothesized that:

1. Because marginalized groups report higher rates of imposter feelings, Black participants would report higher rates of imposter feelings than White participants.

2. Because of implicit biases, participants would be more likely to be impressed that Black women have achieved success compared to White women.

3. Participants with higher rates of imposter feelings would report more empathy and understanding for the women for whom imposter feelings were explicitly described (imposter condition) than for the women who was not described as feeling like an imposter (control condition).

4. Higher levels of affective or cognitive empathy would predict more positive reactions to the women in the imposter conditions than women in the control conditions.

We had an additional exploratory question. Because of documented tendencies for adults to over-attribute the imposter feeling to marginalized groups, participants might have been more likely to perceive the two Black women as experiencing imposter feelings in comparison to the two White women. However, a competing possibility is that in the condition specifically stating that the woman feels like an imposter, participants might have been more likely to perceive the two imposter condition women as experiencing imposter feelings compared to the two control condition women. In other words, we were interested in whether race and condition would interact, or if one of the two variables (race or imposter status) would prove more salient to young adults.

Method

The study was not preregistered. Three researcher-created items that assessed perceptions of the experimental condition were accidentally miscoded by the researchers and thus were not included in the analyses. These items were similar to other items that were included. All other measures in the study are reported.

Participants

We originally had 121 participants respond to the survey; however, participants who did not correctly respond to our three validity check items were eliminated (see Materials section below). All remaining participants responded to the majority of survey items. Items that were skipped were analyzed as missing data, and no outliers were removed. Thus, our final sample included 113 adults with a mean age of 23.88 (*SD* = 9.22). When we asked participants their identity, 93 were cisgender women, 14 were cisgender men, 1 was transgender, one identified as other, and 4 were non-binary. When we asked participants their sexual orientation, 76 participants identified as heterosexual, 5 as gay/lesbian, 22 as bisexual, 2 as asexual, 4 as pansexual, and 4 as other. When we asked participants to categorize their race, 58 were White, 43 were Black, 6 were Latinx/Hispanic, 3 were Asian, 2 were multi-racial, and 2 was "Other."

This study was approved by our institutional IRB. We recruited participants through personal contact, undergraduate classrooms, Facebook, e-mail, and social networking sites. Potential participants were issued an invitation that included the web address to link to the online survey. Once participants went to that address, they encountered the informed consent form. Choosing to go to the web address was completely voluntary on the part of the participants and required participants to click a link indicating that they agreed to participate.

Materials and Procedure

First, participants responded to the Clance Imposter Phenomenon Score (CIPS; Clance, 1985). Twenty items evaluated the extent to which the participant experienced feelings of fraudulence or unworthiness associated with their accomplishments. A sample item was "I have often succeeded on a test or task even though I was afraid that I would not do well before I undertook the task." Participants made responses on a 5-point Likert scale where 1 represented "not at all true" and 5 represented "very true." The published reliability for this scale ranged from 0.85 to 0.96 (Mak et al., 2019), and we had a Cronbach's alpha of .91.

Next, participants responded to the Basic Empathy Scale (Jolliffe & Farrington, 2006). These 20

items assessed cognitive and affective empathy. Affective empathy refers to the ability to feel what others feel. A sample item was "I tend to feel scared when I am with friends who are afraid." Cognitive empathy refers to the ability to mentally understand the feelings of others. A sample item was "When someone is feeling 'down' I can usually understand how they feel." The average Cronbach's alphas for these scales are .81 (Cabedo-Peris et al., 2021). We had a Cronbach's alpha of .76 for cognitive empathy, and .84 for affective empathy.

We then randomly assigned participants to one of four conditions. The conditions can be seen in the Appendix. In each condition, participants read a brief narrative of a woman who was working in a male-dominated environment where imposter feelings would be likely. For example, all four narratives stated that the character "is currently in a high-ranking position in her career at Company…She is in a male dominated career field, and she is the only woman working at her office." The conditions varied in two ways. First, the woman was described as either Black or White. Second, the conditions differed in how specifically imposter feelings were mentioned. Two conditions stated that the women "feels like an imposter but hides those feelings from everyone around her" and included symptoms of imposter phenomenon (imposter condition). The other two conditions did not contain the word imposter (control condition). Instead, the woman was described as feeling confident and enjoying the challenges of her job. However, the last sentence of these two conditions stated that "It's difficult to work in a male dominated industry but she is making it happen." The created conditions were labeled Black Control, Black Imposter, White Control and White Imposter.

Immediately after reading the story, participants encountered three items that served as a validity check to ensure that participants read the story. The first question asked the participant to recall the name of the company stated in the story. The second asked the participant to choose the answer that best describes the woman they read about. The third asked them to choose the answer that best described the job the woman mentioned.

Next, participants encountered five researcher-created items that measured participants' perceptions of the woman. One item directly assessed our manipulation by stating, "This woman is probably experiencing imposter syndrome." (Note that we did not provide participants with a definition of imposter syndrome.) Two items assessed whether participants perceived the woman positively or negatively: "I am impressed by this woman" and "This woman is probably annoying to work with." We also assessed participants' pity and empathy for the woman. The two items were "I empathize with this woman and her situation" and "I feel sorry for this woman." Participants then encountered an additional researcher-created item that stated, "If I were in this situation, I would experience self-doubt." Responses were made on a 5-point Likert scale where 1 represented "strongly disagree" and 5 represented "strongly agree."

Last, participants encountered five commonly used demographic questions that assessed age, race, gender, and sexual orientation.

Results

Hypothesis One and Participant Race Comparisons

We conducted independent t-tests to compare the responses of participants who self-identified as either Black or White. Compared to White participants, Black participants scored lower on the imposter phenomenon scale (CIPS), $t(99) = 3.71$, $p < .001$, $d = .75$, 95% CI [.34, 1.15]. The mean for White participants was 68.98 (SD = 13.51) and for Black participants was 59.0 (SD = 13.17). Means can be seen in Figure 1. This finding contradicts our hypothesis that Black participants would report higher rates of imposter feelings and reflect a moderate effect. Compared to White participants, Black participants were less likely to agree that they "would doubt themselves in the same situation" as the woman in the scenario, $t(69) = 1.86$, $p = .034$, $d = .46$, CI [-.03, .95]; this reflects a small effect. Compared to White participants, Black participants also scored lower on the affective empathy scale, $t(99) = 2.50$, $p = .007$, $d = .50$, CI [.10, .90], reflecting a moderate effect. We found no

Figure 1
Imposter Syndrome Score of Participants

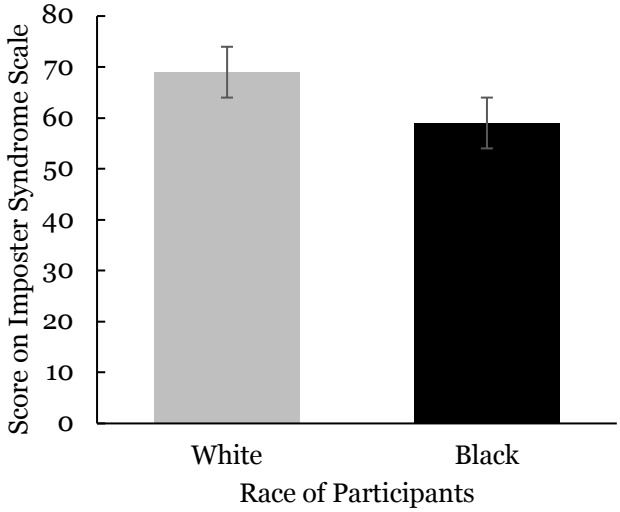


Table 1
Means and Standard Deviations for Dependent Variables Across Race and Imposter Conditions

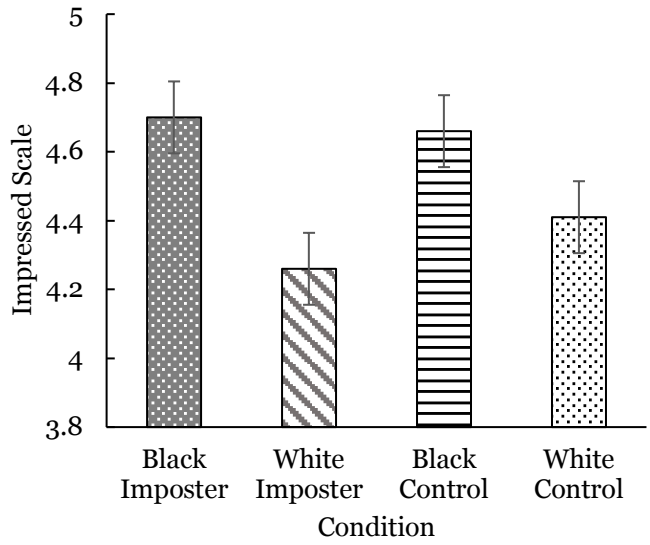
	Imposter Condition		Control Condition	
	Black Woman	White Woman	Black Woman	White Woman
Impressed By	4.70 (.54)	4.26 (.92)	4.66 (.55)	4.41 (.95)
Feel Sorry For	3.37 (1.18)	2.65 (1.34)	2.25 (1.24)	2.07 (1.22)
Empathize With	4.33 (.68)	4.04 (1.26)	3.78 (1.04)	3.90 (1.21)
Annoying to Work With	1.33 (.68)	1.65 (1.03)	1.41 (.76)	1.66 (1.01)
Experiencing Imposter	3.44 (1.09)	4.04 (1.04)	3.06 (1.41)	2.97 (1.40)

differences between Black and White participants' scores on the cognitive empathy scale [$t(99) = .81$, $p = .21$, *ns*].

Hypothesis Two and Condition Comparisons

We compared the responses to the four researcher-created items that assessed participants' perceptions of the described woman using a 2(race) X 2(imposter status) MANOVA. The means and standard deviations can be seen in Table 1. We found no significant interactions. In support of our hypothesis, there was a main effect for race, where participants agreed more that they were "impressed by this woman" for the Black, versus White conditions, $F(1, 111) = 5.70$, $p = .019$, $\eta 2 = .05$. This represents a small effect. See Figure 2 for the means. There was a main effect for the imposter condition, with participants agreeing more that they felt "sorry for this woman" for the imposter, versus the control, conditions, $F(1, 111) = 12.84$, $p < .001$ $\eta 2 = .03$. This represents a small effect. We also found a main effect for the imposter condition, where participants who read the imposter condition scenarios were more likely to agree that the woman was "probably feeling imposter syndrome," $F(1, 111) = 9.26$, $p = .003$ $\eta 2 = .09$. This represents a medium effect. We found no main effects for how much participants "empathize with this woman and her situation" [$F(1, 111) = .18$, $p = .669$, *ns*] or for how much participants agreed that the "woman is probably annoying to work with" [$F(1, 111) = 2.89$, $p = .092$, *ns*].

Figure 2
Means for Participants Impressed by Different Conditions



Hypothesis Three and Participants' Imposter Phenomenon Scores as Predictors

Because we found no interactions, we split the data file between the four conditions to examine relations among variables. Participants with higher CIPS agreed more that they would "doubt themselves in the same situation" when they read about the Black Control woman [$r(21) = .66$, $p = .001$] and Black Imposter woman [$r(23) = .55$, $p = .006$]. These reflect moderate to strong correlations. Participants with higher CIPS also agreed more that they "empathize with" [$r(27) = .41$, $p = .035$] and felt "sorry" [$r(27) = .41$, $p = .035$] for the Black Imposter woman. These reflect moderate correlations. Participants' CIPS did not predict their perceptions of the White women. This analysis did not support our hypothesis that participants with higher rates of imposter feelings would report more empathy and understanding for the Imposter, versus the Control, woman.

Hypothesis Four and Participants' Empathy Scores as Predictors

Our examination of empathy did not support our hypothesis that higher levels of empathy would predict more positive reactions to the Imposter, versus the Control, woman. Instead, the race of the target woman was more influential. Higher scores on the affective empathy scale correlated with participants agreeing more that were impressed with the White Control woman [$r(30) = .47$, $p = .010$] and White Imposter woman [$r(24) = .52$, $p = .010$]. These reflect moderate correlations. The higher participants' affective empathy the less

they agreed that the White Control woman [$r(29) = -.45$, $p = .014$] and White Imposter woman [$r(24) = -.52$, $p = .009$] were "probably annoying to work with." These reflect moderate correlations. Participants' affective empathy levels did not predict their perceptions of the Black women.

The higher participants' cognitive empathy, the more they felt empathy for the Black Control woman [$r(32) = .35$, $p = .047$], Black Imposter woman [$r(27) = .51$, $p = .006$], and White Control woman. [$r(30) = .44$, $p = .015$]. These reflect weak to moderate correlations. Cognitive empathy did not predict empathy for the White Imposter woman. Instead, participants with higher cognitive empathy responded to the White Imposter woman by agreeing more that they "were impressed" [$r(24) = .72$, $p < .001$] and agreeing less that she was "probably annoying to work with" [$r(24) = -.68$, $p < .001$]. These are strong correlations.

Discussion

Hypothesis One: Imposter Feelings by Participant Race

We hypothesized that, in line with previous research, our Black participants would report higher rates of imposter feelings than our White participants. This hypothesis was not supported. In fact, Black participants in our study had lower imposter scores and disagreed that they would doubt themselves if they were in the imposter situation described in our narratives. Our sample was drawn largely from young adults with some college education, and our scenario described someone who was advanced in her career. College students may not be able to predict how much doubt they would feel in a situation that they had not yet encountered. However, this argument does not explain why Black, versus White, participants reported less imposter feelings (a clear contradiction with previous research). One possibility is that, because of the age of our participants, they experienced some portion of their schooling during the isolation of COVID-19; the forced separation from peers may have decreased opportunities for experiences of social discrimination and comparison. Social comparison, particularly in a face-to-face context, is known to increase imposter feelings, particularly among women (Fassi et al., 2020; Fraenza, 2016).

Most of our Black participants were recruited from a racially diverse campus, which may also have reduced their experiences with social discrimination, social comparison, and thus, imposter feelings. In support of this argument, Bernard et al. (2024) recently found that Black college students report higher rates of imposter feelings when their campus racial climate was negative, a link mediated by social anxiety. In other words, perceived campus inequities related to higher anxiety in Black students, and those Black students reported more imposter feelings (Bernard et al., 2024). The racially diverse nature of the campus from which our Black participants were recruited also may have afforded them a certain amount of social support. In systematic literature review, Bravata et al. (2020b) found that increased social support is related to lower levels of imposter feelings. In conjunction, these findings suggest the need for continued investigation into contextual and historical factors that may work alongside other intersectional factors that impact an individual's vulnerability to imposter feelings.

An exploratory question in our research was whether race or imposter status would determine which women, or woman, would be perceived as experiencing imposter feelings. In our study, imposter status emerged as more influential. Participants who read either of the imposter conditions agreed more strongly that the woman was experiencing imposter feelings. They also reported feeling sorry for the two women who were explicitly described as experiencing imposter feelings compared to the two women who were not. This finding provides evidence that our manipulation worked because we described these women as feeling like imposters, and our participants agreed that the women felt like imposters. In other words, our use of the word "imposter" in the narrative description was salient enough to get participants to attribute "imposter syndrome" to the women in that specific condition. (The wording of the scenarios can be seen in the Appendix.) Previous research suggested that adults over-attribute imposter feelings to marginalized groups (Feenstra et al., 2020; McGee et al., 2021), which would have predicted that the Black women in our study would have been seen as "feeling like an imposter;" however, we did not see this in our study. Recall that our Black participants reported lower levels of imposter feelings than our White participants, and thus, perceptions on these items may reflect the personal experiences of our participants.

Hypothesis Two: Impressed Perceptions by Target Race

We hypothesized that participants would be more impressed by the successful Black women compared to the successful White women. This prediction was supported. The women in our scenario were described as succeeding in a male-dominated world and being highly educated. Being impressed by the Black woman may reveal an implicit bias that this level of achievement is less common for minority women and thus, more noteworthy; this implicit bias has been clearly established by previous researchers (e.g., Iheduru-Anderson, 2022;

Pogrebna et al., 2024). Black women may internalize the global perception that Black women's success is uncommon, leading to the higher rates of imposter feelings typically reported for Black women (Noskeau et al., 2021; Wall & Troisi, 2020). An alternative and more optimistic explanation might be that our participants assumed that the workplace was dominated specifically by White men, versus Black men, a perception that would make sense given typical workplace demographics (Afran et al., 2020). If this assumption was made, participants may have been particularly impressed by the Black women because their minority status included both race and gender, whereas White women only represented gender minority status. In other words, the Black women might have been perceived as impressive because they achieved workplace success despite membership in two historically disadvantaged groups in the workplace (Dill & Duffy, 2022; Pogrebna et al., 2024).

Hypothesis Three: Imposter Feelings Predict Empathy and Understanding

We hypothesized that increased personal feelings related to believing one was an imposter would predict greater empathy and understanding for the two women in the conditions where imposter feelings were clearly described. This hypothesis was not supported. Participants with more personal feelings related to believing they were an imposter were more likely to relate to the Black, versus the White women, regardless of her imposter condition. These participants, high in imposter feelings, also were more likely to agree that they empathized with and felt sorry specifically for the Black Imposter woman. In other words, participants' own imposter feelings predicted their responses to the women's race more than to the women's imposter experience. Perhaps adults who are aware of imposter feelings assumed that all Black women experience imposter feelings, and because they personally know that imposter feelings are unpleasant, they responded with empathy and pity. Such an assumption might be fair because Black women are more likely to report imposter feelings (Chrousos & Mentis, 2020; Fields & Cunningham-Williams, 2021). Another possibility is that the result reveals a microaggression. White participants in our study were more likely to experience imposter syndrome – perhaps they revealed an implicit bias by assuming if they, personally, were feeling like an imposter, a Black woman would be even more likely to feel like an imposter. This argument would be in line with previous researchers who documented that people over-attribute the imposter label to minority women (Cawcutt et al., 2021; Feenstra et al., 2020; McGee et al., 2021; Simons, 2021).

Hypothesis Four: Empathy Score and Perceptions of Conditions

We also hypothesized that higher levels of affective or cognitive empathy would predict more positive reactions to the women in the imposter conditions than women in the control conditions. This hypothesis was not supported, and like our earlier presented findings, the race of the woman was more predictive of perceptions than her imposter status. Participants who were higher in affective empathy (e.g., more capable of feeling what others feel) were more impressed with the White woman and believed her to be more less annoying to have as a co-worker. One possible explanation is that this outcome was driven by our White participants. White participants scored higher on affective empathy, and it might be these same White participants who preferred the White woman in the workplace. In other words, White adults may be showing a bias for their own race, a bias that exists across race and culture (Meissner & Brigham, 2001; Wong et al., 2020). A similar argument might be that White adults, who were adept at feeling what others feel, had an especially easy time identifying and connecting with the emotions of White people.

When we examined cognitive empathy (e.g., the ability to understand another's situation) we found a somewhat similar pattern. Participants higher in cognitive empathy reported feeling more empathy for all the women except the White Imposter. Instead of feeling empathy for her, they were impressed by her and believed her to be less annoying. This pattern is close to what we saw with affective empathy; however, affective empathy predicted positive feelings for both White women, while cognitive empathy predicted positive feelings specifically for the White Imposter woman. People high in cognitive empathy analyze situations with logic in addition to emotion. The data suggest that there was something about the combination of "White" and the inclusion of the word "imposter" in the scenario that led these participants to be particularly impressed with the woman. Maybe these two terms (White, imposter) seemed less compatible than Black and imposter, leading to deeper processing and reflecting some implicit bias.

Limitations and Conclusion

The main limitation of this study is the lack of diversity within the sample. There was diversity in race. However, our sample had a disproportionately high percentage of women (87%) compared to men. This gender disparity might be especially relevant given that our narratives focused on women characters. In addition to broadening participant demographics, future researchers may want to collect more data about adults'

own experiences at work or in school to start identifying specific experiential factors that may drive perceptions. Future researchers could also focus on how perceptions translate toward behaviors. For example, does having empathy for someone experiencing imposter feelings impact interactions with that individual?

In sum, our data did not clearly support our hypotheses. Compared to Black participants, White participants reported higher levels of imposter feelings. Similarly contradicting previous research, our participants did not assume that our Black women were more likely to be experiencing imposter feelings. Although we expected our imposter condition to influence perceptions, we found stronger patterns related to race. Our participants were more impressed with the Black women who was succeeding in a male-dominated workplace than with the White women in the same circumstance. Adults who experienced imposter feelings themselves were more understanding of Black women, while adults higher in empathy, who were disproportionately White women, reported more positive perceptions of the White women. These findings suggest that imposter feelings may not always be higher, or perceived to be higher, in Black, compared to White, women. In addition, our findings add to the limited research examining how perceiver characteristics impact attributions of imposter feelings by demonstrating that perceivers' empathy and personal experiences with imposter feelings influence their perceptions.

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Appendix

Experimental Conditions

Imposter Condition	A Black/White woman is currently in a high-ranking position in her career at Company AI. She has spent years working hard to earn 2 degrees that have qualified her for the position. She is the first person in her family to graduate from college, and they are very proud of her. She is in a male dominated career field, and she is the only woman working at her office. She is having trouble accepting her accomplishments, wondering if she deserves the praise that she is getting. She worries whether she can keep doing as good of a job in the future as she has done in the past and whether she will ultimately disappoint her family. Often, she feels like an imposter but hides those feelings from everyone around her.
Control Condition	A Black/White woman is currently in a high-ranking position in her career at Company CD. She has spent years working hard to earn 2 degrees that have qualified her for the position. She is the first person in her family to graduate from college, and they are very proud of her. She is in a male dominated career field, and she is the only woman working at her office. She feels confident in her ability and enjoys the praise she gets from others. She has to work hard but she enjoys the challenge. She feels especially proud of herself when she faces a tough situation and work and figures out a solution to the problem. It's difficult to work in a male dominated industry but she is making it happen.